# Chapter 3. Random variables

<u>Def'n</u>: A random variable is a function that assigns a real number to each outcome in S.

* $X$ = R.V. ( temperature )

* $x$ = number. ( 70, 80...)

<u>Def'n</u>: A discrete R.V. can assume a finite or countably infinite number of values.

<u>Def'n</u>: A continuous R.V. can assume any value in an interval.

<u>Example:</u> Flip a coin twice.

$S = \{HH, HT, TH, TT\}$

Let $X$ = # of H.

a> what is the "support of X"

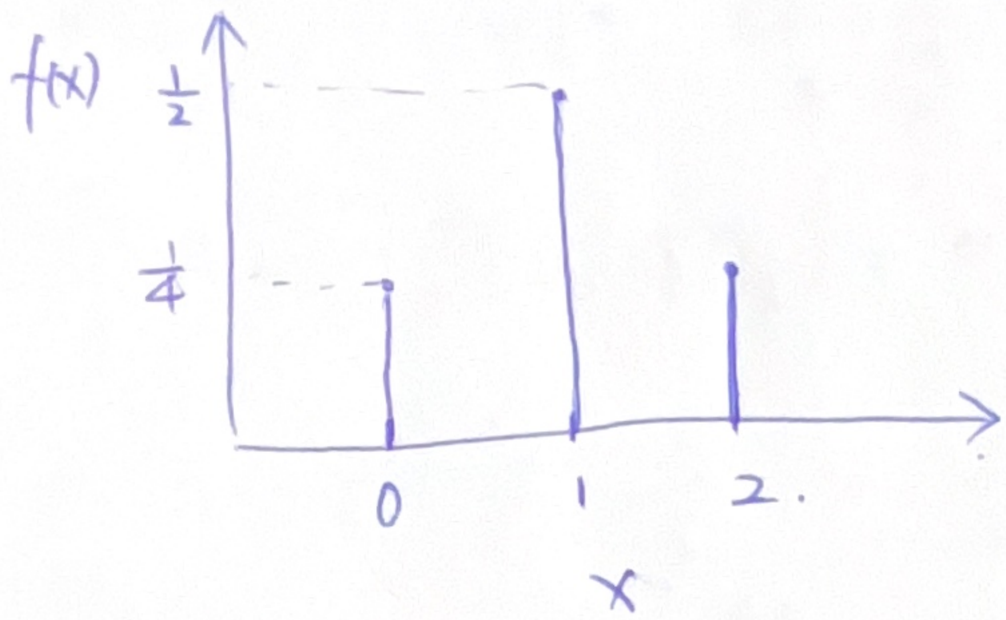
$S_X = \{0, 1, 2\}$

b>. Find the <u>probability mass function</u> of X

pmf

| $x$ | 0 | 1 | 2 |
|---|---|---|---|
| $f(x) = P(X=x)$ | $\frac{1}{4}$ | $\frac{1}{2}$ | $\frac{1}{4}$ |

$$f(x) = \begin{cases} \frac{1}{4} & \text{for } x=0 \\ \frac{1}{2} & \text{for } x=1 \\ \frac{1}{4} & \text{for } x=2. \end{cases}$$




<u>Notes</u>: Suppose $X$ is a discrete R.V.

1) $f(x) > 0 \quad \forall x \in S_x$

2) $\sum_{x \in S_x} f(x) = 1.$

$$\sum_{x \in S_x} f(x) = f(0) + f(1) + f(2) = 1.$$

3) $P(X = x) = f(x)$

Example: Bowl has 6 chips

2 chips → "2" 4 chips → "4"

R.S. 2 chips <u>without</u> replacement.

Let $X$ = sum of the 2 chips drawn.

$S_x = \{4, 6, 8\}$

a) Find pmf of $X$

| $x$ | 4 | 6 | 8 |
|---|---|---|---|
| $f(x)$ | $\frac{1}{15}$ | $\frac{8}{15}$ | $\frac{6}{15}$ |

$$f(x=4) = P(X=4) = \frac{\#\text{"4"}}{\#S} = \frac{\binom{2}{2}}{\binom{6}{2}} = \frac{1}{15}.$$

$$P(X=4) = P(2_1 \cap 2_2) = P(2_1)\,P(2_2 \mid 2_1)$$

$$= \frac{2}{6} \cdot \frac{1}{5} = \frac{1}{15}.$$

$$P(X=6) = \frac{\binom{2}{1}\binom{4}{1}}{\binom{6}{2}} = \frac{8}{15}.$$

$$P(X=6) = P[(2_1 \cap 4_2) \cup (4_1 \cap 2_2)]$$

$$\overset{me}{=} P(2_1 \cap 4_2) + P(4_1 \cap 2_2)$$

$$= P(2_1)\,P(4_2 \mid 2_1) + P(4_1)\,P(2_2 \mid 4_1)$$

$$= \frac{2}{6} \times \frac{4}{5} + \frac{4}{6} \times \frac{2}{5}$$

$$= \frac{8}{15}.$$

$$P(X=8) = \frac{\binom{4}{2}}{\binom{6}{2}} = \frac{6}{15} = \frac{2}{5}$$

b) $$P(X \geq 6 \mid X \leq 6) = \frac{P(X \geq 6 \cap X \leq 6)}{P(X \leq 6)} = \frac{P(X=6)}{P(X \leq 6)}$$

$$= \frac{\frac{8}{15}}{\frac{1}{15} + \frac{8}{15}} = \frac{8}{9}.$$

c). Let $X$ = product of 2 chips drawn.

Find pmf of $X$

| $x$ | 4 | 8 | 16 |
|---|---|---|---|
| $f(x)$ | $\frac{1}{15}$ | $\frac{8}{15}$ | $\frac{6}{15}$ |

d). $P(X=6) = 0$.

<u>Cumulative distribution function</u> cdf. <u>CDF</u>.

<u>Def'n</u>: cdf of a discrete R.V is

$$F(x) = P(X \le x) = \sum_{x_i \le x} f(x_i)$$

where. 1) $\lim_{x \to -\infty} F(x) = 0$

2) $\lim_{x \to \infty} F(x) = 1$.

3) If $x \le y$, then $F(x) \le F(y)$